JARVIS CONTINUAL-LEARNING GOVERNANCE

PHASE 8: MODEL EVALUATION & SHADOW PROMOTION FLOW — FINAL ACCEPTANCE REPORT

Date: September 17, 2026
Environment: Controlled Non-Prod

PHASE 8: CLOSED / ACCEPTED | **LEVEL: NON-PRODUCTION HARDENING** | **PRODUCTION: STRICTLY DISABLED**

Executive Summary & Official Decision

Phase 8 has been officially accepted and closed under the agreed non-production operational-hardening designation. All 25 operational invariants (`INV-SHD-001` through `INV-ART-002`) and 50 dedicated acceptance scenarios (`P8-001` through `P8-050`) have been verified with 100% pass rates across 6 modular test suites.

> *"Phase 8 is officially CLOSED / ACCEPTED under the agreed non-production operational-hardening designation. Operational hardening has been implemented and verified in the controlled test environment. Real-hardware, cross-session, cross-reboot, power-loss, and production-readiness validation remain pending on detached external validation tracks."*

Formal Status Breakdown

```
PHASE_8: IMPLEMENTATION COMPLETE
STATUS: UNIT_VERIFIED + INTEGRATION_VERIFIED + OPERATIONALLY_HARDENED_IN_TEST_ENVIRONMENT

DEDICATED_ACCEPTANCE_SCENARIOS: 50/50 PASSED
LEARNING_REGRESSION_SUITE: 446/446 PASSED
DESKTOP_HOST_SUITE: 34/34 PASSED

REAL_HARDWARE_VERIFIED: PARTIAL/PENDING
REAL_GPU_VERIFIED: NO — SIMULATED
REAL_TOOL_SANDBOX_VERIFIED: CONTROLLED-ENVIRONMENT VERIFIED; BARE-METAL OS ENFORCEMENT PENDING
CROSS_PROCESS_VERIFIED: CONTROLLED TEST VERIFIED
CROSS_SESSION_VERIFIED: NO
CROSS_REBOOT_VERIFIED: NO
POWER_LOSS_RECOVERY_VERIFIED: NO

CRASH_RECOVERY_VERIFIED: VERIFIED_IN_SUITE
ACTIVATION_ATOMICITY: VERIFIED_IN_SUITE
ROLLBACK_RECOVERY: VERIFIED_IN_SUITE
SAFE_DEGRADED_MODE: VERIFIED_IN_SUITE
PROMPT_INJECTION_ISOLATION: VERIFIED_IN_SUITE
FALLBACK_METRIC_CONTAMINATION: VERIFIED_IN_SUITE

LIVE_TRAINING: STRICTLY DISABLED
AUTOMATIC_MODEL_PROMOTION: STRICTLY DISABLED
AUTOMATIC_MODEL_DEPLOYMENT: STRICTLY DISABLED
AUTOMATIC_STRATEGY_PROMOTION: STRICTLY DISABLED
PRODUCTION_ACTIVATION: STRICTLY DISABLED
OPERATOR_APPROVAL: STRICTLY REQUIRED

PRODUCTION_CANDIDATE: NO
PRODUCTION_ACTIVE: NO
```

Separation of Verified Scope vs. Pending External Tracks

Subsystem / Capability Area	Verified Interpretation	Status Classification
50 Phase 8 Scenarios (P8-001 to P8-050)	All passed cleanly in modular test suites in 1.87s	**PASSED (50/50)**
Learning Core Regression Suite	All historical regression tests passing cleanly	**PASSED (446/446)**
Desktop Host Interaction Suite	HookBus integration and tool routing verified	**PASSED (34/34)**
Shadow Side-Effect Isolation	Dry-run simulator intercepts mutations (WhatsApp, shell, files)	**CONTROLLED VERIFIED**
Cross-Process Execution	Isolated subprocess dispatch and worker crash containment	**CONTROLLED VERIFIED**
Activation Transaction Atomicity	Two-phase commit (pending → commit → verify → audit)	**IN-SUITE VERIFIED**
Rollback & Safe Degraded Mode	Hysteresis, known-good target guard, SAFE_DEGRADED_MODE	**IN-SUITE VERIFIED**
Bare-Metal Hardware & Real GPU	Simulated in controlled environment; physical GPU lab pending	**PENDING TRACK**
Cross-Session & Cross-Reboot Recovery	Reboot and AC power-loss recovery pending external lab	**PENDING TRACK**
Production Activation & Auto-Promotion	Autonomous promotion strictly prohibited by governance locks	**STRICTLY DISABLED**

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PHASE 8: TECHNICAL HARDENING ARCHITECTURE & IMMEDIATE ACTIONS

Reference: INV-SHD-001 to INV-ART-002

Key Technical Hardening Architecture

1. Cryptographically Bound Operator Approval Token (INV-AUTH-001)

Operator promotion authorization tokens bind human identity, execution context, and cryptographic digests:

```
OperatorApprovalToken(
    operator_identity="operator_alice",
    operator_os_sid="S-1-5-21-3623811015-3361044348-30300820-1013",
    session_id="sess_auth_987",
    operator_role="learning_admin",
    model_id="cand_v100",
    artifact_digest="sha256_art_...",
    checkpoint_digest="sha256_ckp_...",
    evaluation_report_digest="sha256_rep_...",
    runtime_contract_digest="sha256_contract_...",
    approval_nonce="nonce_single_use_456",
    approval_timestamp=1789665000.0,
    expiry_timestamp=1789668600.0,
    token_signature="hmac_sha256(...)"
)
```

2. Measured Service-Level Objectives for Shadow Dispatch (INV-SHD-004)

Universal absolute latency claims replaced by measured statistical SLO distributions:

- **Shadow Dispatch Overhead:** p50 ≤ 1.0ms, p95 ≤ 5.0ms, p99 ≤ 20.0ms hard ceiling.
- **Active-Path Added Latency:** ≤ 1.0ms maximum allowable overhead.
- **Zero Active Request Blocking:** On queue saturation, shadow requests shed non-blockingly without delaying active turns.

3. Explicit 7-Tier Data Sensitivity Classification (INV-PRIV-001)

Granular classification ensures even read-only observations cannot leak sensitive user activity:

- `PUBLIC_READ_ONLY`: Time, volume, window title (safe for shadow read).
- `SENSITIVE_READ_ONLY`: Browser tabs, app state, clipboard (sanitized snapshot only).
- `PRIVILEGED_READ_ONLY`: Browser cookies, credentials, private keys (strictly denied in shadow).
- `SIMULATABLE`: Focus window, switch tab, set volume (dry-run simulated state).
- `SIDE_EFFECTING`: WhatsApp, delete file, write file, execute shell (intercepted, never executed).
- `FORBIDDEN`: Direct socket writes, process injection (hard blocked at gateway boundary).
- `REDACTED` / `UNAVAILABLE`: Stripped tokens and masked credentials.

4. Stale-Result Protection & Task Generation Tracking (INV-VOICE-001, INV-CORR-001)

Every active turn carries `conversation_id`, `turn_id`, `task_generation`, `cancellation_token`, and `state_version`:

- Barge-in voice cancellation terminates pending execution within 150ms.
- Commit guards abort stale commits if cancellation token was signaled before state mutation.
- User corrections bump `task_generation`; actions belonging to older generations are discarded immediately.

5. Strict Fallback Metric Schema Isolation (INV-SHD-002)

Metric schema explicitly separates candidate performance from safety fallbacks:

- `candidate_direct_success`: Evaluated solely on turns where candidate executed directly.
- `fallback_success`: Turns completed by fallback LLMs are strictly excluded from candidate accuracy.

Immediate Actions & Phase 9 Roadmap

1. **Lock Phase 8 Governance:** All production promotions, auto-deployments, and live learning remain strictly disabled.

2. **Maintain Controlled Environment:** Maintain all 446 regression tests, 50 Phase 8 scenarios, and 34 desktop interaction tests green.
3. **Establish Detached Hardware Lab Track:** Schedule bare-metal GPU, multi-reboot persistence, and AC power-loss recovery tests for external tracking.
4. **Transition to Phase 9:** End-to-end live desktop & browser execution, cross-session reflection persistence, and truthful state evidence export.